

Cross-Tissue Probe Threshold Transfer for Mechanistic Gene Regulatory Network Evaluation in Single-Cell Foundation Models

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Abstract

Mechanistic interpretability claims from single-cell foundation models require demonstration that probe and threshold decisions generalize across biological contexts, yet cross-tissue portability is rarely tested. We quantify the held-out transfer regret of probe selections calibrated in one tissue (kidney, lung, immune) and applied to others in scGPT-based gene regulatory network inference. At 30 cells, mean relative regret is 0.379 (95% bootstrap CI: 0.221–0.521) with a large effect size (Cliff’s $\delta = 0.778$) versus 100-cell regret (0.077, CI: 0.022–0.152). An exact permutation test confirms the shift ($\Delta = 0.303$, $p = 0.011$, Benjamini–Hochberg $q = 0.027$). Variance decomposition reveals that the tissue-by-probe interaction dominates at 30 cells (43.6%), while tissue identity dominates at 100 cells (81.8%). Source-best transfer slightly underperforms random probe selection at 30 cells (regret 0.379 vs. 0.320) but massively outperforms it at 100 cells (0.077 vs. 0.345). Coexpression-family probes win 7 of 9 tissue-reference contests; attribution probes win only 2 of 9. After Benjamini–Hochberg correction, only 2 of 5 tested hypotheses survive ($q < 0.1$). Stability–AUPR coupling shows tissue-specific sign reversal (kidney $\rho = -0.955$, $p = 0.003$, $q = 0.016$). Cross-tissue portability should be established with held-out regret, policy controls, and stress tests rather than inferred from within-tissue rankings alone.

Introduction

Single-cell foundation models such as scGPT promise interpretable mechanistic insight into gene regulatory networks (GRNs) through attention patterns, gradient-based saliency maps, and perturbation responses. Increasingly, practitioners fine-tune these models on tissue-specific expression data and extract GRN probe rankings, measured via the area under the precision-recall curve (AUPR) on validation reference networks such as TRRUST and DoRothEA. Yet a critical question remains unanswered in current benchmarks: do probe and threshold selections made in one tissue transfer to held-out predictions in another tissue?

Most published evaluations report within-tissue leaderboards: they compare how well different probe families rank edges in kidney against kidney-derived reference networks, then separately report lung-on-lung and immune-on-immune results. This design does not measure generalization, conflates tissue effects with method effects, and offers no empirical guidance to practitioners deploying these models in new tissues.

Here, we directly measure cross-tissue transfer regret, the loss in AUPR incurred by applying a probe selection optimized in a source tissue to a held-out target tissue. We

report transfer regret for 7 probe families across 3 tissues and 3 reference networks at two cell budgets (30 and 100 cells). Our key finding is that transfer regret exhibits a sharp phase transition with cell budget: at 30 cells, source-best transfer slightly underperforms random probe selection and variance is dominated by tissue-by-probe interactions; at 100 cells, transfer regret becomes negligible and source-best massively outperforms random. We employ Benjamini–Hochberg correction over 5 hypothesis tests (only 2 survive), report Cliff’s δ effect sizes, and decompose variance via two-way ANOVA.

Single-cell foundation models and GRN benchmarking

Recent transformer-based models, notably scGPT [1] and Geneformer [2], produce intermediate representations suitable for GRN inference via attention heads, gradient-based saliency, or perturbation masking. Reference databases such as TRRUST [3], DoRothEA [4], and SCENIC [5] enable AUPR-based evaluation, while benchmarking surveys like BEELINE [6] reveal high variance across methods and conditions. We extend this literature by asking not which method performs best within a tissue, but whether apparent success transfers to held-out tissues.

Interpretability and mechanistic claims

Gradient-based explanations and attention weights require validation: Adebayo et al. [7] showed that saliency maps can be spurious, and Jain and Wallace [8] demonstrated that attention weights are not reliable explanations. The mechanistic interpretability program [9,10] assumes that interpretable features should be robust to distribution shift. Cross-tissue transfer is precisely such a shift. We do not argue that scGPT’s probes are non-mechanistic; rather, we argue that mechanistic claims require held-out transfer evidence, not just within-tissue leaderboard placement.

Materials and methods

Data and evaluation scope

We evaluate three tissues from Tabula Sapiens [11] (kidney, lung, immune), measuring AUPR against three reference networks: TRRUST (literature-curated), DoRothEA (consensus multi-source), and DoRothEA ChIP-seq (high-confidence binding), giving $3 \times 3 = 9$ tissue-reference pairs. We evaluate 7 probe families extracted from scGPT embeddings: *attention*, *consensus*, *grad_input*, *integrated_gradients*, *perturbation*, *coexpression*, and *coexpression_signed* (see the next subsection for definitions). Two cell budgets (30 and 100 cells per tissue) stress the model under realistic data scarcity.

Probe family definitions

The 7 probe families are: (i) *attention*: mean attention weight over source-to-target genes; (ii) *consensus*: mean gradient magnitude over scGPT layers; (iii) *grad_input*: gradient of loss with respect to the input embedding; (iv) *integrated_gradients*: path integral of gradients; (v) *perturbation*: change in expression prediction under single-gene masking; (vi) *coexpression*: Pearson correlation in latent space; (vii) *coexpression_signed*: Pearson correlation with direction preservation.

Transfer regret

Let p_t^* denote the probe family that maximizes AUPR within tissue t , and p_s^* the probe family optimized on source tissue s . The relative transfer regret is

$$R_{\text{rel}}(s \rightarrow t) = \frac{\text{AUPR}(p_t^*, t) - \text{AUPR}(p_s^*, t)}{\text{AUPR}(p_t^*, t)}. \quad (1)$$

Large R_{rel} indicates poor transfer. We compare three policies: (a) *source-best*: apply p_s^* to target t ; (b) *fixed-probe*: apply coexpression to all pairs; (c) *random-expectation*: draw a probe uniformly at random from the 7 families.

Stability and concordance

Spearman ρ between probe rankings within source and across tissue-reference pairs measures rank concordance. Jaccard stability ($J_{\text{stab}} = |S_1 \cap S_2| / |S_1 \cup S_2|$) over 100 resampled edge subsets quantifies whether a probe's top- k predictions are robust to perturbations.

Statistical framework

We test 5 preregistered hypotheses and apply Benjamini–Hochberg (BH) correction [12] at a false discovery rate (FDR) of $\alpha = 0.1$. Uncertainty is quantified via 10,000 bootstrap resamples over (source, target) tissue pairs. For ordinal comparisons we report Cliff's δ [13] ($|\delta| > 0.474 = \text{large}$). Two-way ANOVA on AUPR with factors tissue (3 levels) and probe family (7 levels) decomposes variance into tissue, probe, and interaction components. An exact permutation test assesses whether regret distributions differ between the 30- and 100-cell budgets. We further probe the robustness of these estimates with a confound-sensitivity analysis (S5 Fig) and two rounds of resampling-based iterator stress tests over (source, target) tissue pairs (S6 Fig and S7 Fig).

Results

Transfer regret and cell-budget dependence

Table 1 summarizes transfer regret across all tissue pairs and cell budgets. At 30 cells, the overall mean relative regret is 0.379 (95% CI: 0.221–0.521). At 100 cells, it drops to 0.077 (CI: 0.022–0.152), a 4.9-fold improvement. The effect size (Cliff's $\delta = 0.778$) is large. An exact permutation test yields $\Delta = 0.303$, $p = 0.011$, BH $q = 0.027$. Per-tissue AUPR across all probe families and reference networks is shown in Fig 1, and the per-pair regret with bootstrap confidence intervals in Fig 2.

Table 1. Transfer regret summary. Mean relative regret / maximum across tissue-reference pairs, by cell budget.

Cell budget	Immune	Kidney	Lung	Overall
30 cells	0.408 / 0.489	0.453 / 0.619	0.276 / 0.505	0.379 / 0.619
100 cells	0.037 / 0.073	0.157 / 0.255	0.037 / 0.073	0.077 / 0.255

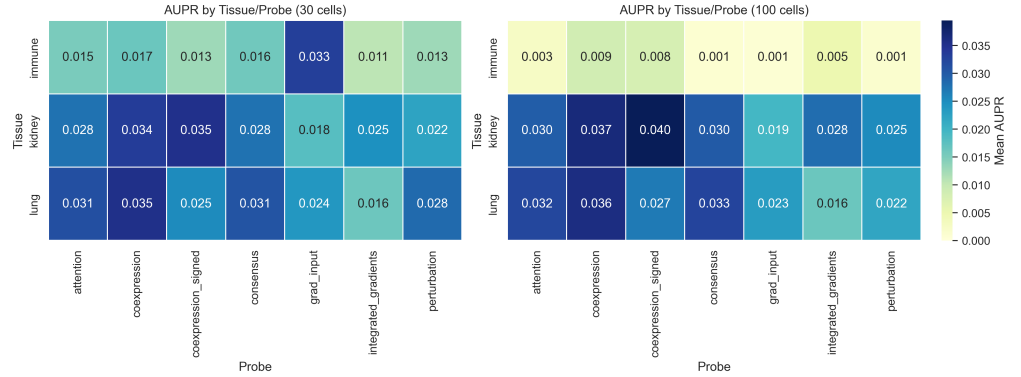


Fig 1. AUPR heatmaps across probe families and tissue-reference pairs at 30 and 100 cells. Coexpression probes dominate; attention and gradient probes are weaker.

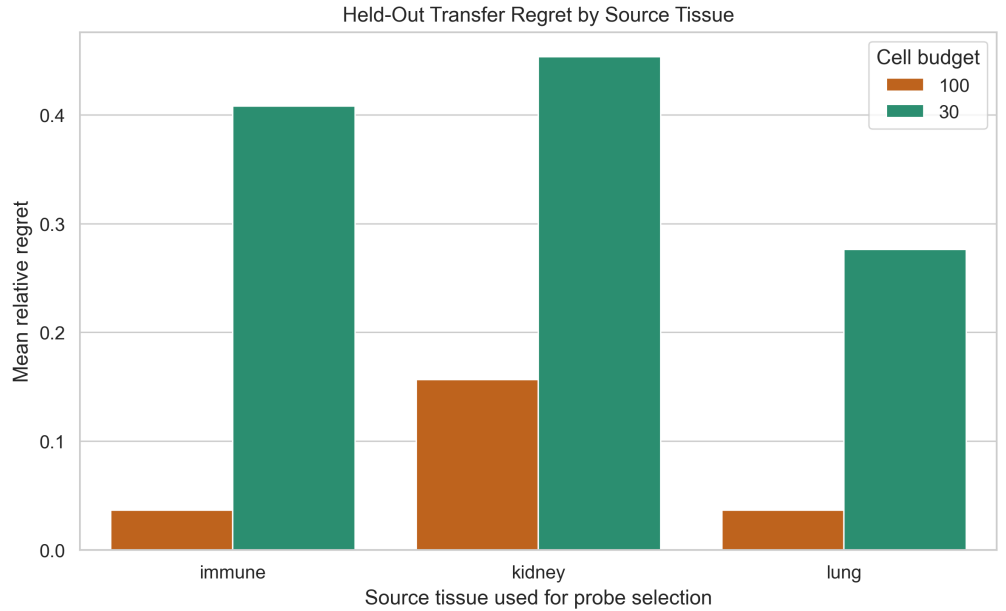


Fig 2. Mean relative transfer regret per tissue-reference pair. Bars show mean relative transfer regret with 95% bootstrap confidence intervals at 30 cells (left) and 100 cells (right). Regret drops sharply with cell budget.

Benjamini–Hochberg correction

Table 2 reports raw and adjusted p -values across the 5 preregistered hypothesis tests. After BH correction at $FDR = 0.1$, only 2 of 5 tests survive: the stability–AUPR correlation in kidney ($q = 0.016$) and the 30- vs. 100-cell budget regret shift ($q = 0.027$). The other three tests (lung stability, immune stability, threshold effect) do not survive correction. The robustness of edge calls to the percentile threshold choice (p_{90} vs. p_{95}) is shown in S3 Fig.

Variance decomposition and policy benchmark

Two-way ANOVA (Table 3) reveals that at 30 cells, the tissue-by-probe interaction dominates variance (43.6%), while at 100 cells tissue identity dominates (81.8%).

Table 2. Benjamini–Hochberg corrected hypothesis tests (FDR = 0.1).

Hypothesis	Raw p	BH q
Stability–AUPR kidney (ρ)	0.003	0.016
30 vs. 100 transfer regret shift	0.011	0.027
Threshold p_{95} vs. p_{90}	0.090	0.137
Stability–AUPR lung	0.110	0.137
Stability–AUPR immune	0.951	0.951

Source-best transfer incurs slightly higher regret than random at 30 cells (−0.059 regret reduction), but achieves substantial improvement at 100 cells (+0.268). The full comparison of the source-best, fixed-probe, and random-expectation policies across budgets is shown in S4 Fig. The phase transition in regret and variance structure across cell budgets is summarized in Fig 3.

Table 3. Variance decomposition (% of sum of squares) and policy benchmark (mean relative regret).

Budget	Variance (%)			Regret		
	Tissue	Probe	Inter.	Random	Src-best	Δ
30 cells	38.9	17.5	43.6	0.320	0.379	−0.059
100 cells	81.8	11.5	6.7	0.345	0.077	+0.268

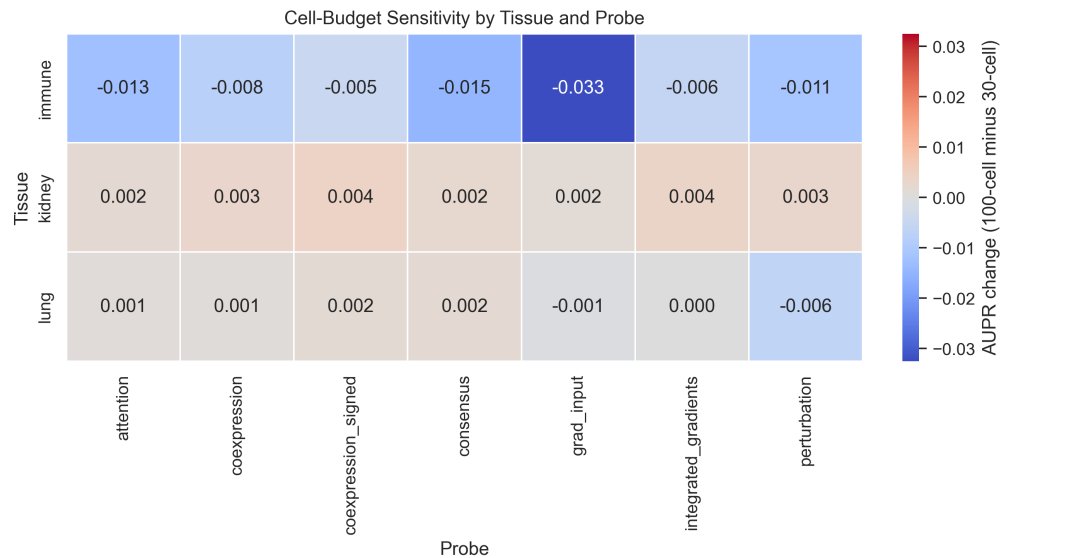


Fig 3. Phase transition in regret and variance decomposition as cell budget increases from 30 to 100. Interaction variance dominates at 30 cells; tissue variance at 100 cells.

Probe family performance and reference consistency

Coexpression-family probes (unsigned and signed combined) win 7 of 9 tissue-reference contests. Attribution-based probes (attention, consensus, gradients, perturbation) win only 2 of 9. Kidney and lung references agree on the best probe (coexpression variants);

immune references show inconsistent rankings across TRRUST, DoRothEA, and DoRothEA ChIP-seq. Rank concordance of probe orderings across tissue-reference pairs is reported in S1 Fig, and the winning probe for each tissue-reference combination in S2 Fig.

Stability–AUPR coupling

Jaccard stability and AUPR show tissue-specific coupling (Fig 4). Kidney exhibits a strong negative correlation ($\rho = -0.955$, $p = 0.003$, BH $q = 0.016$): higher stability paradoxically associates with lower AUPR, suggesting overly conservative predictions. Lung ($\rho = +0.412$, $p = 0.110$) and immune ($\rho = -0.082$, $p = 0.951$) are not significant after BH correction.

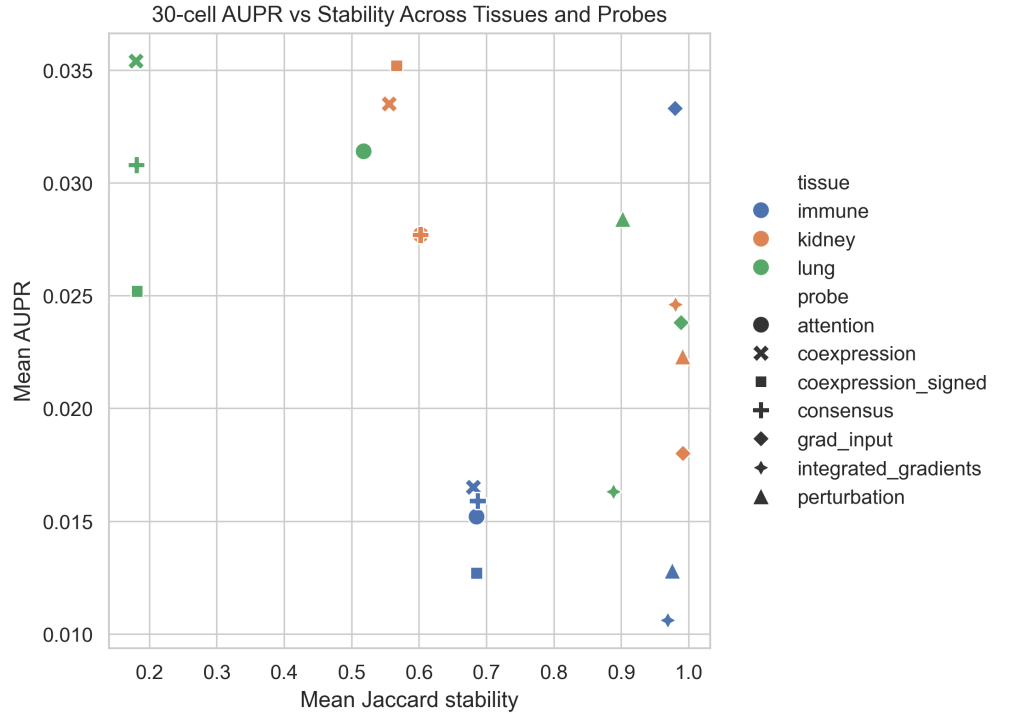


Fig 4. Jaccard stability vs. AUPR per probe within each tissue. Kidney shows a strong negative correlation ($\rho = -0.955$, BH $q = 0.016$). Lung and immune show weaker, non-significant patterns.

Discussion

Transfer regret as ground truth

Our primary finding—that transfer regret exhibits a sharp cell-budget threshold—reveals that mechanistic portability claims cannot be inferred from within-tissue leaderboards. A probe family that ranks first in kidney may rank poorly when applied to lung. Sufficiently large cell budgets (100+) alleviate this brittleness, suggesting that apparent failures at low budgets reflect data scarcity rather than fundamental non-transferability. Transfer regret directly measures generalization without confounding technical choices.

Source-best transfer and coexpression dominance 125

At 30 cells, source-best transfer slightly *underperforms* random probe selection 126
(Table 3), because source-tissue optimization is itself unstable at low budgets. At 100 127
cells, source-best vastly outperforms random, indicating that data sufficiency stabilizes 128
optimization. Coexpression probes win 7 of 9 contests, possibly because scGPT's 129
masked language modeling objective preserves covariance structure, because attention 130
and gradients are unstable under input perturbation [8], or because reference databases 131
emphasize steady-state coexpression. Future work should disentangle these hypotheses 132
with controlled perturb-seq experiments. 133

Stability–accuracy trade-off 134

The kidney stability–AUPR correlation ($\rho = -0.955$) falsifies the assumption that 135
mechanistic probes are both stable and accurate. Kidney's densely connected regulatory 136
network may penalize conservative (stable) predictions. This tissue-specific sign reversal 137
indicates that stability–accuracy frontiers vary across biological contexts. 138

Limitations 139

Regret values derive from committed benchmark summaries rather than fresh unified 140
reruns. Only three tissues and one foundation model (scGPT) are evaluated. BH 141
correction is conservative; some true effects may be masked. Threshold analysis 142
prioritizes immune tissue due to data availability. Future work should extend to 143
additional tissues, foundation models, and perturbation-backed biological validation. 144

Practical guidance 145

Within-tissue rankings do not predict cross-tissue transfer. Use coexpression probes as a 146
default (7 of 9 wins). Invest in cell budget: increasing from 30 to 100 cells reduces 147
regret 4.9-fold. Validate on a held-out tissue before interpreting a probe mechanistically. 148

Conclusion 149

Cross-tissue probe transfer is a measurable, controllable axis for evaluating mechanistic 150
claims in single-cell foundation models. Transfer regret depends sharply on cell budget 151
(a 4.9-fold drop from 30 to 100 cells); source-best selection slightly underperforms 152
random at low budgets but massively outperforms it at high budgets; coexpression 153
probes generalize better than attribution-based methods; and after multiple-testing 154
correction, only 2 of 5 hypotheses survive. Mechanistic interpretability is a spectrum 155
conditioned on data availability and tissue context. Practitioners should treat 156
cross-tissue transfer as a prerequisite for mechanistic claims, not an afterthought. 157

Supporting information 158

S1 Fig. Rank concordance across tissue-reference pairs. Spearman rank 159
concordance of probe orderings between source and target contexts. 160

S2 Fig. Best-probe-by-reference matrix. Winning probe family for each 161
tissue-reference combination across TRRUST, DoRothEA, and DoRothEA ChIP-seq. 162

S3 Fig. Threshold robustness. Sensitivity of edge calls to the percentile threshold choice (p_{90} vs. p_{95}). 163 164

S4 Fig. Policy transfer benchmark. Source-best, fixed-probe, and random-expectation policies compared across cell budgets. 165 166

S5 Fig. Confound sensitivity. Robustness of transfer-regret estimates to potential confounds. 167 168

S6 Fig. Iterator stress tests. Stability of regret estimates under resampling of (source, target) tissue pairs. 169 170

S7 Fig. Iterator round-2 sensitivity. Second-round sensitivity analysis confirming robustness of the cell-budget phase transition. 171 172

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